

AY: 2025-2026

Course: PPOM 2

Week 9, Lecture 85

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Clinical: Electrolyte Disorders, Potassium Disorders

Presented by:

Matthew Heller, D.O., FAOASM

Associate Professor, Family and Sports Medicine

NYIT College of Osteopathic Medicine

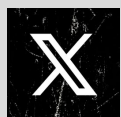
Team Physician, NYIT Athletics

Ringside Physician

Email: mheller@nyit.edu



[@matthewbheller](https://www.instagram.com/matthewbheller)



[@matthewbheller](https://twitter.com/matthewbheller)



Session Objectives

- Differentiate between normal and abnormal potassium values
- Demonstrate an understanding of the diagnosis and treatment of hypokalemia and hyperkalemia
- Formulate and assess a differential diagnosis for patients with hypokalemia and hyperkalemia
- Identify key findings on an EKG related to both hyperkalemia and hypokalemia
- Recognize and Describe how potassium distribution is affected by hypo insulinemic states, beta-blocker administration, insulin administration, and GI potassium binders

Heller's Cookie Model for the Treatments of Hyper- AND Hypo- Kalemia

With Special Guest....



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RED WOLVES and BEARS

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Basics of Lab Results

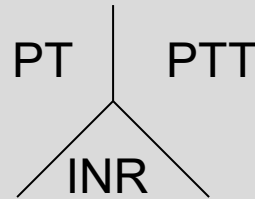
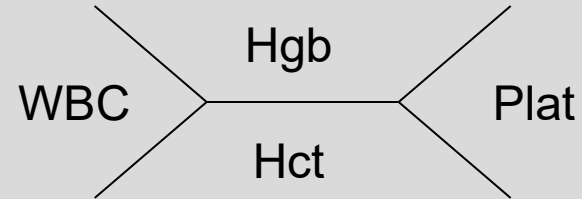
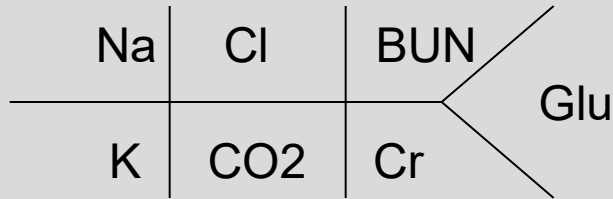
- “The Result is only as good as the specimen provided”
- Location of the draw
 - Avoid drawing from a limb with an IV
 - If necessary, stop IV for 30 minutes and draw venous blood distal to the IV site
- Decrease the risk of hemolysis
 - Use as large bore needle as possible
 - Recognizing a suboptimal blood draw

Basics of Laboratory Medicine involved with Electrolytes disordered

- Sources of Specimens:
 - Venous blood
 - Arterial blood
 - Urine
- Electrocardiogram (EKG) can also be of significance in many cases

How to document

- List or diagram format – Most important is to be consistent

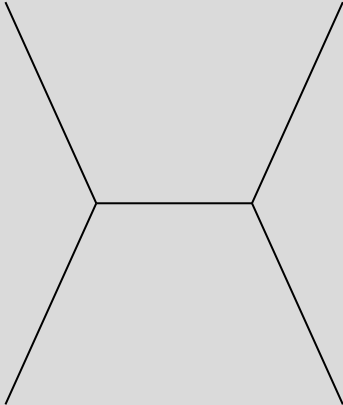
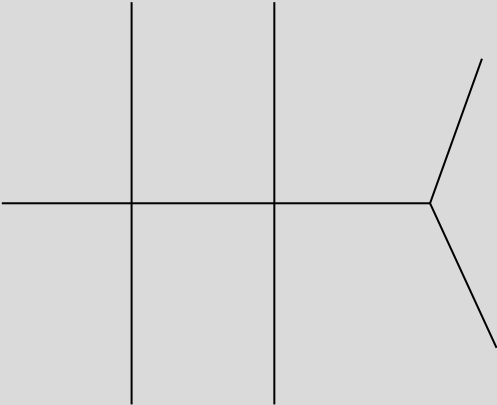


Practicing Documenting Labs

Doe, John -123456
2/27/19 8:30am

ALT	10	8-20 U/L
AST	15	25-125 U/L
T. Bil	1.0	0.1-1.0 mg/dL
D. Bil	0.2	0.0-0.3 mg/dL
Creatinine	0.8	0.6-1.2 mg/dL
Na	138	136-145 mEq/L
Potassium	3.5	3.5-5.0 mEq/L
Chloride	100	95-105 mEq/L
Bicarbonate	25	22-28 mEq/L
Magnesium	1.8	1.5-2.0 mEq/L
Glucose	90	70-110 mg/dL
BUN	12	7-18 mg/dL

WBC	5.0	4.5-11.0 x10 ⁹ /L
Hematocrit	41%	0.41-0.53
Hemoglobin	13.5	13.5-17.5 g/dL
Platelet	250	150-400 x 10 ⁹ /L



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Why is Potassium Important

- Makes up normal osmotic and electrical gradients in the tissue
- Precise transcellular movement of potassium is important for excitable tissues like neuronal transmission, cardiac conduction, excitation contraction coupling of muscles
- It is the “other ion” for acid-base buffering for acute acidosis and alkalosis
- Required for intracellular glucose metabolism, oxidative phosphorylation and protein synthesis
- A constant cell K concentration is critical for enzyme activities and for cell division and growth

How do we control K homeostasis

- ingest
- 100mEq/ day

- distribute
- Intracellular/ extracellular

- Excrete 90-95mEq/day
- Renal 90% (GI 10 %, Sweat < 5 mEq/d)

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Banana
451mg of potassium
12mEq of potassium

TABLE 1

Potassium-Rich Foods

Food group	Food item	Food group	Food item
Seasonings/ condiments	■ Low-sodium salt (contains potassium chloride) ■ Sea salt	Vegetables	■ Seaweed ■ Spinach ■ Tomatoes ■ Broccoli ■ Potatoes
Desserts	■ Molasses ■ Chocolate	Meats	■ Beef ■ Chicken ■ Pork ■ Lamb
Nuts, seeds, and legumes	■ Lima beans ■ Sunflower seeds ■ Pistachios ■ Pumpkin seeds	Fruit	■ Bananas ■ Cantaloupes ■ Oranges ■ Avocados ■ Dried fruits (dates, prunes)
Grains	■ Wheat germ ■ Bran ■ Oats		

■ Highest content (> 1,000 mg per serving)

■ Very-high content (> 600 mg per serving)

■ High content (> 250 mg per serving)

Information from references 12, 16, and 17.

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World Health
Organization
recommends
3,510 mg per day
for adults for
optimal
cardiovascular
health

Distribution of Potassium

ECF 350 mEq (10%)

- Plasma 15 mEq (0.4%)
- Interstitial fluid 35 mEq (1%)
- Bone 300 mEq (8.6%)
- **[K+] = 3.5–5.0 mEq/L**

ICF 3150 mEq (90%)

- Muscle 2650 mEq (76%)
- Liver 250 mEq (7%)
- Erythrocytes 250 mEq (7%)
- **[K+] = 140–150 mEq/L**



Factors causing Transcellular K Shifts

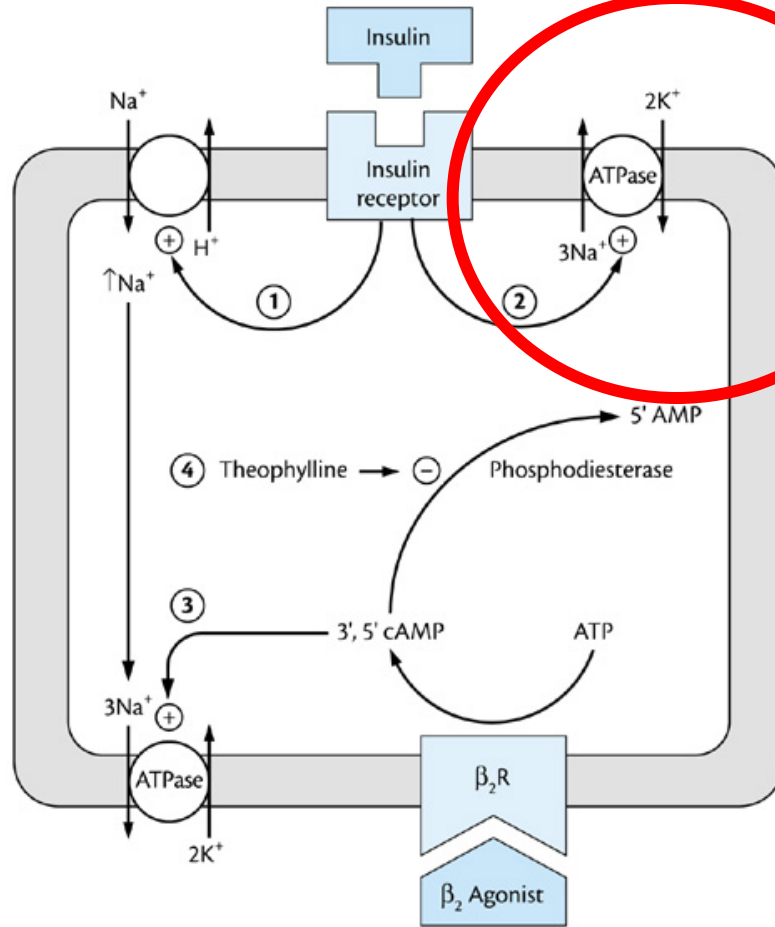
- Acid-base status
 - Metabolic acidosis, Respiratory Acidosis
 - Metabolic alkalosis, Respiratory alkalosis
- Pancreatic hormones
 - Insulin
 - Glucagon
- Catecholamines
 - Beta-Adrenergic
 - Alpha-Adrenergic
- Hyperosmolarity
- Aldosterone
- Exercise

Potassium (K^+) – Renal Excretion

- ▶ Assuming a typical 100mmol intake and normal renal function:
 - Filtered $K^+ = GFR \times [K^+_{\text{serum}}]$
 - Passive reabsorption in the proximal convoluted tubule and Loop of Henle = 90%
 - Active secretion in distal convoluted tubule and the cortical collecting duct by *Principal cells* and *Intercalated cells*
 - Secretion/reabsorption is increased or decreased based on potassium concentrations and aldosterone

Potassium Disorders

- Although the extracellular fluid (ECF) contains about 10% of total body potassium, less than 1% is located in the Plasma.
- Thus, disorders of potassium metabolism can be classified as those that are due
 - 1) to altered intake,
 - 2) to altered elimination,
 - 3) to deranged transcellular potassium shifts.



Hyperkalemia

- Hyperkalemia is made by getting the result from the lab
 - $>5\text{mmol/L}$
- Associated labs/tests:
 - BUN/Creatinine
 - Serum glucose
 - Urine: Potassium and creatinine
 - EKG

Hyperkalemia - Etiology

- Sample error
- Potassium shift
- Decreased excretion
- Excessive potassium intake

Hyperkalemia – Etiology (Potassium Shift)

- Rhabdomyolysis
- Hemolysis
- Burns
- Strenuous exercise
- Sepsis
- Hypertonicity
 - K follows the flow
- Insulin deficiency
- Metabolic acidosis
- Pharmaceuticals
 - Digoxin
 - Beta-antagonists
 - Succinylcholine
 - Arginine

Exercise and Potassium

- During Exercise
 - Skeletal muscle loses K^+ during repetitive action potentials
 - Skeletal muscle is major reservoir for K^+ in body
 - Potassium level may increase markedly to around 8
 - Conditioning reduces the increase in K^+ during exercise
- After Exercise
 - Muscles regain loss potassium by Na/K^+ pump mediated potassium uptake
 - Normalization of potassium within minutes with a possibility for a transient hypokalemia
- Effects on heart
 - Exposed to high potassium during exercise
 - Exposed to major drop in potassium after exercise
 - Drop seems to be associated with impaired cardiac repolarization
 - Could potentially trigger arrhythmia and sudden cardiac death

Keld Per Kjeldsen, Thomas Andersen Schmidt, Potassium homeostasis and pathophysiology of hyperkalaemia, *European Heart Journal Supplements*, Volume 21, Issue Supplement_A, 1 February 2019, Pages A2–A5, <https://doi.org/10.1093/eurheartj/suy033>

Hyperkalemia – Etiology (Decreased Excretion)

- Renal failure
- Interstitial nephritis
- Sickle cell disease
- Hypoaldosteronism
 - Type IV RTA
 - Diabetic nephropathy
 - Heparin
 - End stage AIDS
 - Adrenal insufficiency
- Pharmaceuticals
 - ACE inhibitors
 - Trimethoprim
 - NSAIDs
 - Spironolactone
 - Triamterene
 - Pentamidine

Hyperkalemia – Etiology (Excessive Intake)

- Ingestion
 - Dietary-induced hyperkalemia usually involves concurrent renal insufficiency
- Iatrogenic

Hyperkalemia - History

- Immediate danger of hyperkalemia is its effect on cardiac conduction and muscle strength
- Absence of symptoms does not exclude severe hyperkalemia
- More severe cases
 - Muscle weakness
 - Heart Palpitation
 - Ascending paralysis
 - Paresthesia

Hyperkalemia - History

- Population at increased risk
 - Chronic Kidney disease
 - Diabetes
 - Heart Failure
 - Liver disease

Hyperkalemia – Physical Signs

- Normal exam
- Generalized weakness
- Decreased deep tendon reflexes
- Decreased / absent bowel sounds

Hyperkalemia – Approach

- Review patients' medication
- Ask about the use of salt substitutes containing potassium
- Obtain Blood pressure
- Assess for kidney hypoperfusion

Hyperkalemia – Approach

- Repeat Potassium level
- Additional labs
 - BUN/Creatinine
 - Glucose
 - Urine electrolytes and creatinine
- EKG

Hyperkalemia – Approach

- EKG should be considered if
 - $K > 6 \text{ mEq/L}$
 - Symptoms of hyperkalemia
 - Suspicion for rapid-onset hyperkalemia
 - Underlying kidney disease, heart disease, or cirrhosis with a new case of hyperkalemia

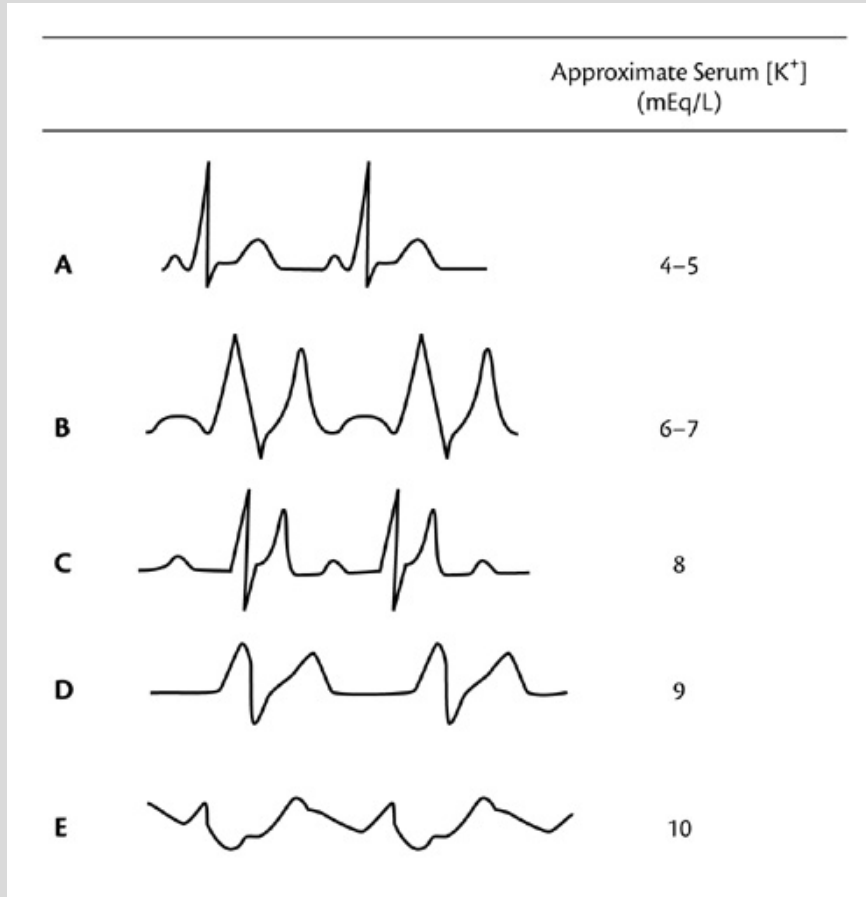
Hyperkalemia – EKG Changes

- Peaked T waves generally the earliest EKG sign
- P-wave flattening
- PR-interval prolongation
- Widening of QRS complex
- Hyperkalemia induced arrhythmias
 - Sinus bradycardia
 - Sinus arrest
 - Ventricular tachycardia / Ventricular fibrillation
 - Asystole

Hyperkalemia – EKG Changes

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Osorio F, Linas S. Atlas of Diseases of the Kidney: Disorders of Water, Electrolytes, and Acid-Base. Edited by Robert Schrier (series editor), Tomas Berl. ©1999

Hyperkalemia – Treatments

- Ultimately, finding the cause and correcting the cause or compensating for it is the long term goal!
- However, due to importance in action potentiation, it requires **rapid treatment** to improve the potassium.
 - If hyperkalemia is suspected, do not delay treatment to find the cause

Hyperkalemia – Treatments

- Goals of acute treatment
 - Prevent potentially life-threatening cardiac conduction
 - Prevent neuromuscular disturbances
 - Shift Potassium into cells
 - Eliminate excess potassium
 - Resolve the underlying disturbance

Hyperkalemia – Treatments

- Indications for prompt intervention
 - Symptoms of hyperkalemia
 - Changes on EKG
 - Severe hyperkalemia ($K > 6.5$)
 - Rapid-onset hyperkalemia
 - Underlying heart disease, cirrhosis, or kidney disease

Hyperkalemia – Treatments

- Intravenous Calcium
 - Stabilizes cardiac muscle cell membrane
 - No effect on serum K^+ or total body K^+
- Insulin and Glucose
 - Shifts K^+ into cells
 - No effect on total body K^+
- Inhaled Beta Agonists
 - Shifts K^+ into cells
 - No effect on total body K^+
- Sodium Bicarbonate
 - Should not be used as monotherapy
 - May have a role among patients with metabolic acidosis
- Hemodialysis
 - The method of choice for removal of potassium from the body in patients with kidney failure

Intravenous Calcium

- When you have too many cookies, you need to balance it out with Milk (CALCIUM) to create balance!
- Cookies taste good with milk!

Source: Cookie Monster



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Stabilizes cardiac
muscle cell membrane

No effect on serum K⁺
or total body K⁺

Inhaled Beta Agonist AND Insulin (with dextrose)

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- When you have too many cookies (potassium) in the jar (blood), you can put them in small bags (cells).
- You're not losing any cookies, you're just displacing them (maybe for later?)



Shifts K^+ into
cells

No effect on
total body K^+

Hemodialysis

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- When you have too many cookies, and need to eliminate some altogether by directly taking it out of the blood



Removal of potassium from the body



Method of choice for those with Kidney failure

Source: Cookie Monster

Hyperkalemia – Subacute Treatments

- Kayexalate (sodium polystyrene sulfonate)
 - Binds K^+ in exchange for sodium
 - Lowers total body K^+
 - Association with GI complications
 - Should be avoided in patients at risk of abnormal bowel function

Hyperkalemia – Additional Treatments

Gastrointestinal cation exchanger

- Patiromer (Veltassa)
 - 2015
 - Not for life-threatening hyperkalemia
 - Powder suspension in water for oral administration
 - Binds potassium in the lumen of the GI tract
 - Increase fecal potassium excretion
- Sodium Zirconium Cyclosilicate (Lokelma)
 - Shown to reduce serum potassium by -0.67 mEq per L at 48 hours
 - Faster decrease in serum potassium at one compared to Patiromer

Hyperkalemia – Additional Treatments

- Furosemide
 - Renal secretion of potassium
 - Effect dependent on urine output
 - Use with caution in patients who are hypovolemic or with renal impairment

Kayexalate (Sodium Polystyrene Sulfonate)

Preferred- Patiromer (Veltassa)

Preferred- Sodium Zirconium cyclosilicate (Lokelma)

Kim MJ, Valerio C, Knobloch GK. Potassium Disorders: Hypokalemia and Hyperkalemia. Am Fam Physician. 2023 Jan;107(1):59-70.

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- When you have too many cookies, you can take Kayexalate to help potassium get eliminated through your GI Tract.
- You can DO this in front of the TV at home, so it's not for a life-threatening situation
- Be careful for a belly ache!!



Source: Cookie Monster

Lowers total body K+

Non-renal elimination

Given Oral or Rectally
(Kayexalate only)

Exchanges sodium for
potassium in the GI TRACT
and increases the fecal
excretion of potassium

Associated with GI
Complications

Kayexalate (Sodium Polystyrene Sulfonate)

Preferred- Patiromer (Veltassa)

Preferred- Sodium Zirconium cyclosilicate (Lokelma)

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Source: Source: <https://giphy.com/gifs/sesamestreet-xT0xeMA62E1XIlup68>

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Lowers total body K+

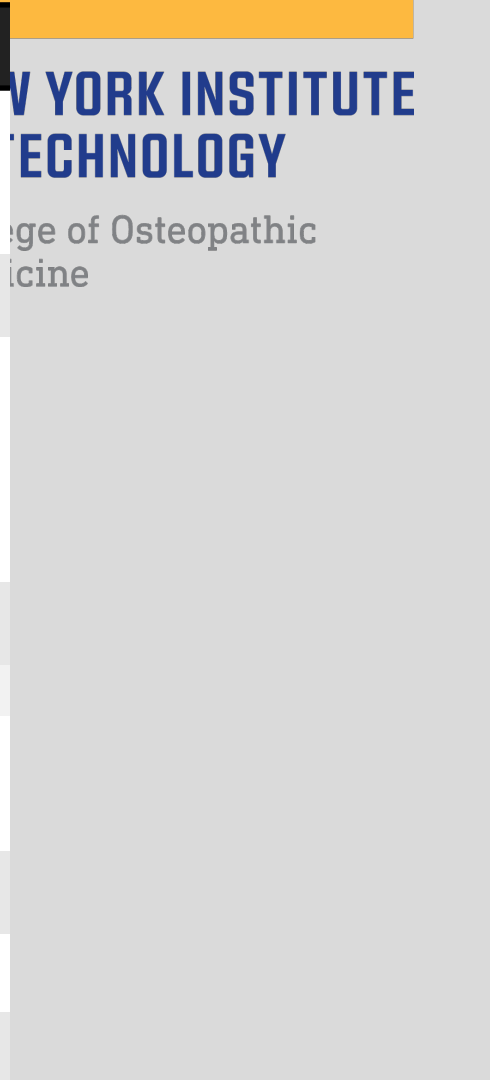
Non-renal elimination

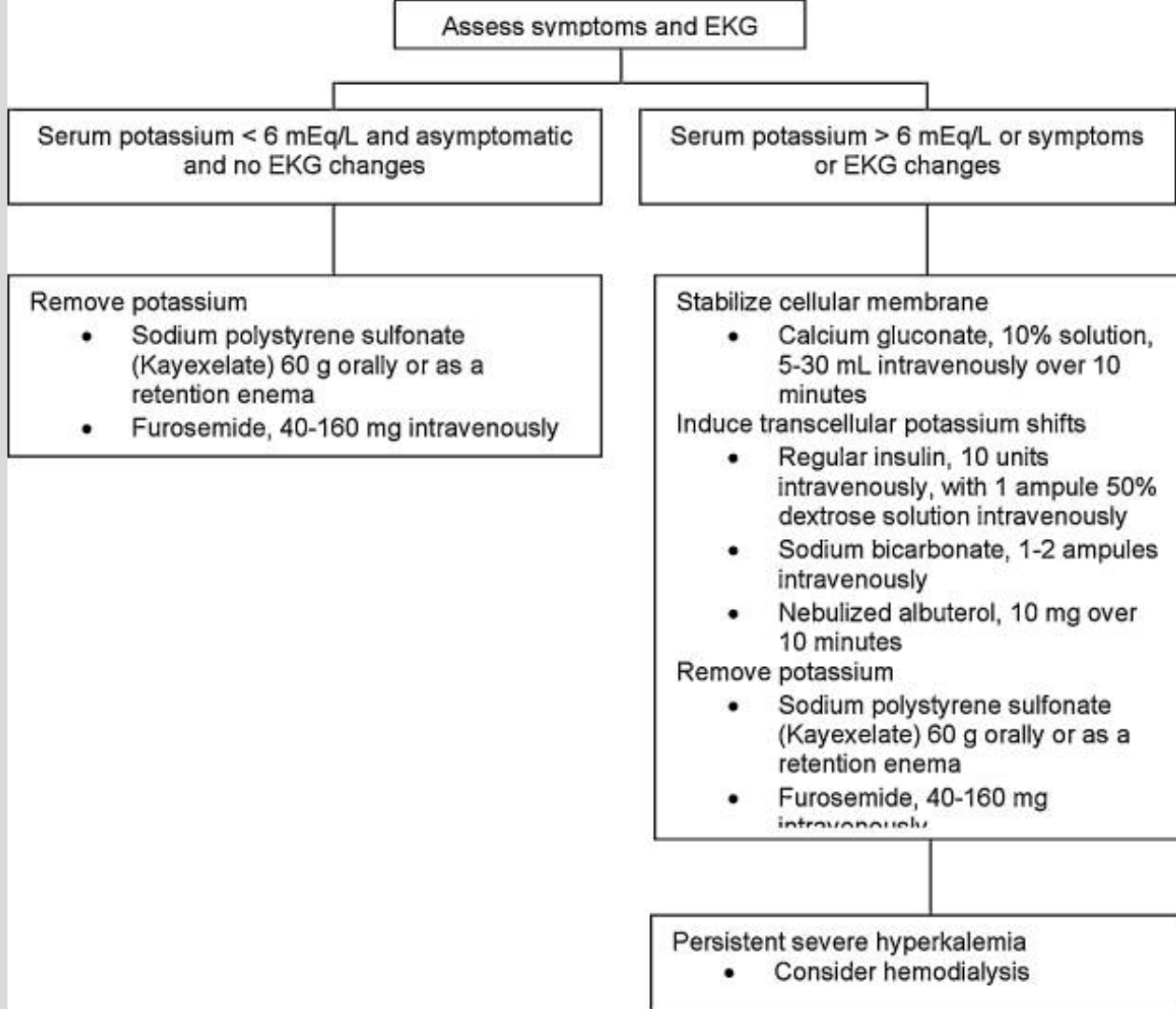
Given Oral or Rectally
(Kayelatate only)

Exchanges sodium for
potassium in the GI TRACT
and increases the fecal
excretion of potassium

Associated with GI
Complications

Treatment	Usual dose	Route of administration	Onset/duration	Expected result
Calcium gluconate	1 g (4.4 mEq)	IV over 5 min	1-2/10-30 min	Reversal of cardiac toxicity
Calcium chloride	1 g (13.5 mEq)	Central IV over 5-10 min	1-2/10-30 min	(Note: no effect on K ⁺ levels)
Insulin (regular)	1 unit/3-5 g dextrose	IV bolus or infusion	10-20 min/2-6 h	Shift of K ⁺ intracellularly
Dextrose	50 mL D50W (25 g)	IV over 5 min	30 min/2-6 h	Shift of K ⁺ intracellularly
	0.5-1.0 L D10W	IV over 1-2 h		Maintains blood glucose
				Avoid if hyperglycemic
Salbutamol/albuterol 0.5%	10-20 mg	Nebulized over 10 min (diluted)	30 min/1-2 h	Shift of K ⁺ intracellularly
				Not as monotherapy
Sodium bicarbonate	50-100 mEq	IV bolus or infusion	30-60 min/2-6 h	Shift of K ⁺ intracellularly Effects unreliable
Sodium polystyrene sulfonate	15-60 g	Orally or rectally	1-6 h/variable	Nonrenal elimination of K ⁺
Furosemide	20-80 mg	IV bolus or infusion	5-30 min/2-6 h	Renal elimination of K ⁺
Hemodialysis	—	—	Immediate/variable	Increased elimination of K ⁺





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Brief Management Algorithm for Hyperkalemia

Source: Wilttrout C. Hyperkalemia, Bradycardia, and Cardiac Arrest during Percutaneous Declothing of an Arteriovenous Graft. Semin Intervent Radiol. 2010 Jun;27(2):241-4. doi: 10.1055/s-0030-1253520. PubMed PMID: 21629412; PubMed Central PMCID: PMC3036522.

Hyperkalemia – Treatment - Chronic

- Counseling to reduce dietary potassium
- Adjustment in medications
- Avoiding NSAIDs
- Adding a diuretic if the patient has sufficient renal function

HYPOkalemia

Hypokalemia

- Occurs in up to 2.9-7.4% of hospitalized patients
- Potassium levels less than 3.5 mEq/L
- Etiology
 - Decreased intake
 - Renal potassium loss
 - Potassium shift favoring intracellular
 - Decreased potassium intake
 - Typically coupled with another cause
 - Extrarenal potassium loss
 - Sample error

Hypokalemia – Etiology (Renal Potassium Loss)

- Increased aldosterone effect
 - Hyperaldosteronism
 - Renovascular HTN
 - Cushing's Syndrome
 - Licorice
 - Congenital Anomaly
 - 11-beta or 17-alpha hydroxylase deficiency
- Increased flow at distal nephron
 - Diuretics
 - Furosemide
 - Thiazide diuretics
- Hypomagnesemia
- Renal Tubular Acidosis
 - Fanconi Syndrome
 - Interstitial Nephritis
 - Metabolic Alkalosis
- Genetic Disorders
 - Bartter's Syndrome
 - Gitelman's Syndrome
 - Liddle's Syndrome

Hypokalemia – Etiology (Potassium Shift into the Cells)

- Insulin excess
 - Postprandial
 - Iatrogenic cause
- Alkalosis (Respiratory or Metabolic)
- Beta-adrenergic excesses
- Hypokalemic periodic paralysis
 - Familial or Thyrotoxic

Hypokalemia – Etiology (Extrarenal losses)

- Vomiting and/or diarrhea
- Villous adenoma
- Zollinger-Ellison syndrome

Hypokalemia – Symptoms

- None
- Muscle cramps
- Fatigue
- Weakness
- Constipation
- History of Paralysis, hyperthyroidism, or use of insulin or beta agonists

Hypokalemia – Physical findings

- Physical findings
 - Blood pressure
 - Inc with aldosteronism
 - Inc with Liddle's
 - Dec with Gitelmans
 - Dec with diuretic abuse
 - Irregular heart rates
 - Paralysis
 - Decreased bowel sounds

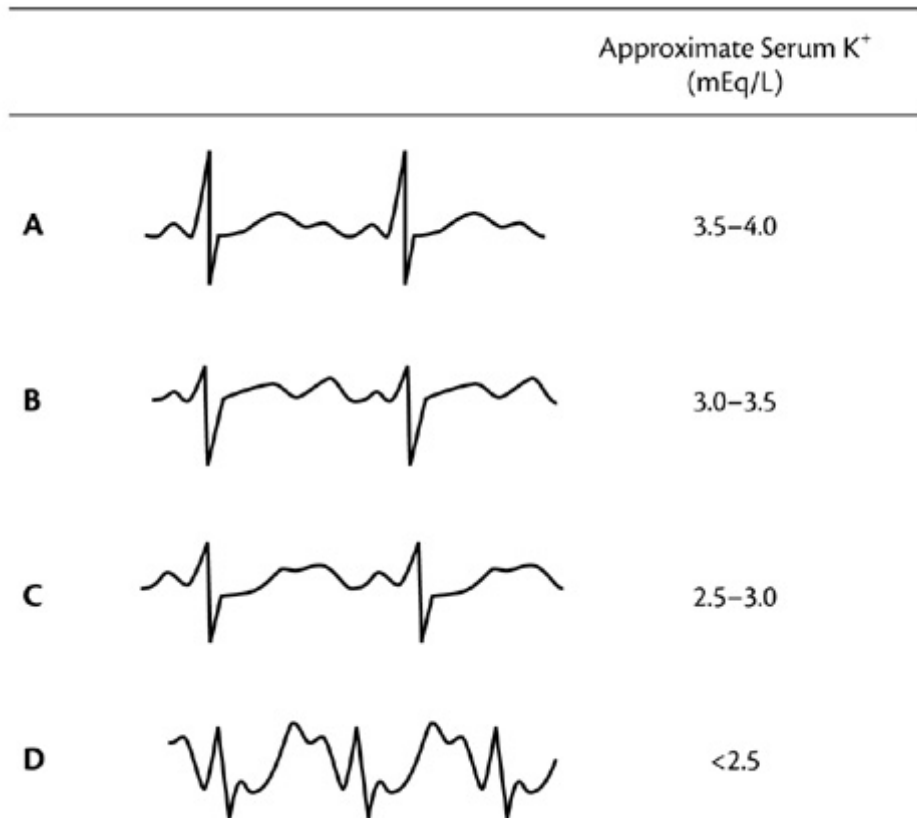
Hypokalemia – Physical findings

- Diagnosis is made by obtaining a lab result finding of $K < 3.5 \text{ mmol/L}$
- Other tests to aid in dx and tx
 - BUN/Creatinine
 - Glucose
 - Magnesium
 - EKG
 - ABG
 - 24 hour urine potassium

Hypokalemia – EKG

- Decreased T-wave amplitude
- With further progression
 - ST-interval Depression
 - T-wave Inversions
 - PR-interval prolongation
 - U-waves
 - Arrhythmias
 - Sinus Bradycardia
 - Ventricular tachycardia or fibrillation
 - Torsade de pointes

Hypokalemia – EKG Changes



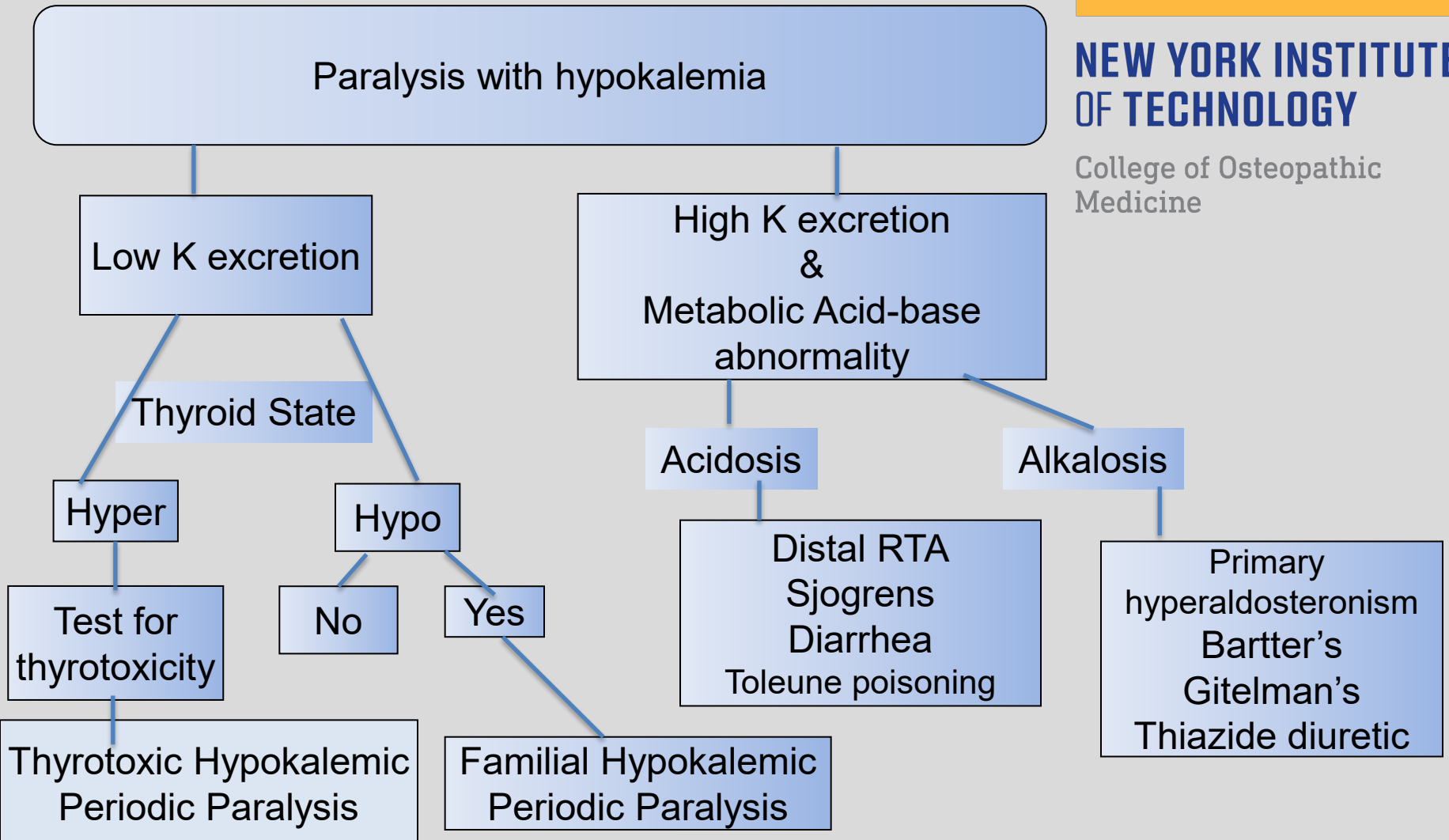
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Osorio F, Linas S. Atlas of Diseases of the
Kidney: Disorders of Water, Electrolytes,
and Acid-Base. Edited by Robert Schrier
(series editor), Tomas Berl. ©1999

Hypokalemia – Urinalysis

- 24 hour urine
 - < 30 mmol = extrarenal cause
 - > 30 mmol = renal cause
- Urine potassium-to-creatinine ratio
 - Higher than 13 meq/g is seen with renal potassium wasting
- Transtubular Potassium Gradient
 - Gauge renal potassium secretion
 - Values > 2 during hypokalemia point to renal loss
 - Values < 6 indicate an inappropriate renal response to hyperkalemia



Thyrotoxic Hypokalemic Periodic Paralysis (TPP) vs. Familial Hypokalemic Periodic Paralysis (FHPP)

	Thyrotoxic Periodic Paralysis	Familial Hypokalemia Periodic Paralysis
Age	20-40	<20
Sex	M>F	Equal distribution
Heredity	No dominance	Autosomal dominant
Ethnicity Predominance	Asian, American Indian, Hispanic, Caucasian	Caucasian, Asian
Hyperthyroidism	Yes	No
Genetic finding	SNPs of Ca _v 1.1	Mutations of Ca _v 1.1

- Candidate for potassium replacement
 - Hypokalemia
 - Patient unable or unwilling to eat low Na diet
 - Patient with nausea, vomiting, diarrhea
 - Patient unable to stop diuretic or laxative
 - Patient with drug-related hypokalemia

Hypokalemia –Treatment

- Prevention of potentially life-threatening cardiac conduction disturbances and neuromuscular dysfunction
- Raise K⁺ to safe level
- Management of underlying disorder
- Patients with a history of CHF or MI should maintain a serum K⁺ of at least 4 mEq per L, based on expert opinion
- Role of Magnesium replacement

Hypokalemia –Treatment

- Determining the approximate potassium deficit
- Serum K^+ drops about 0.3 mEq per L for every 100-mEq reduction in total body potassium
- Example
 - Decline in serum K^+ from 3.8 to 2.9 mEq/L corresponds to about a 300-mEq reduction in total body potassium

Hypokalemia –Treatment

- Obtain a good medical history and list of medications
- If associated with diuretic → stop or reducing the strength
- If has a comorbid condition, may take advantage of
 - ACE inhibitor
 - ARB
 - Beta blocker
 - Potassium sparing diuretic

Hypokalemia –Treatment

- Increase dietary potassium
 - Low-normal and mild hypokalemia
 - Especially in patient with HTN or Heart Disease
 - Effects may be limited
 - Most of the potassium in food coupled with phosphate
 - Most cases of hypokalemia involve chloride depletion and would respond to potassium chloride



Hypokalemia –Treatment

- Intravenous potassium (20-40 mmol in 1 L of NS)
 - Increased risk of hyperkalemia
 - Pain and phlebitis
 - Used in
 - Severe hypokalemia
 - Hypokalemic EKG changes
 - Physical signs or symptoms of hypokalemia
 - Patients unable to tolerate the oral form
- Should not be given in dextrose-containing solutions {Can be given with dextrose, if needed}
- Cardiac Monitoring if rate exceeds 10 mmol per hour

Hypokalemia –Treatment – Non-Urgent

- Oral potassium
 - Non-urgent hypokalemia
 - 40 to 100 mmol per day is usually effective
 - May be used in prevention of hypokalemia in patients with persistent losses
 - Patients on diuretic therapy
 - hyperaldosteronism



Treatment for HypoKalemia

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- When you have too little cookies left in the jar, you need to add more!
- You can either eat more cookies (PO), or you can get it through an IV
- It's more fun to eat the cookies! It can hurt to take it through an IV!

Source: Cookie Monster



Can increase potassium
in your diet

Can increase through
potassium supplements

Can increase through IV
(given with any I.V.
fluids)

Hypokalemia –Treatment

- In emergency situations:
 - Check the electrolytes every 1-2 hours or as conditions change
- In stable situations:
 - Check 4-6 hours after treatment as conditions change
 - Then check daily, or if conditions change
- In Outpatient setting:
 - Often will check weekly until stable then monthly or quarterly depending on patient compliance

Summary Slide

- Interpretation of Abnormal Potassium
- Diagnosis and Treatment of Hypokalemia and hyperkalemia
- Identify key findings on an EKG when potassium too high vs. too low
- Know the treatments and treat the underlying cause

Reference Slide

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- Core References:

- Loscalzo J, Fauci A, Kasper D, Hauser S, Longo D, Jameson J. eds. *Harrison's Principles of Internal Medicine*, 21e. McGraw-Hill Education; 2022. Chapter 53. Subtitle “Potassium Disorders.”
<https://login.nyit.idm.oclc.org/login?url=https://accessmedicine-mhmedical-com.nyit.idm.oclc.org/content.aspx?bookid=3095§ionid=265471636#1197684893>
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